Scrum Framework

Week 3b

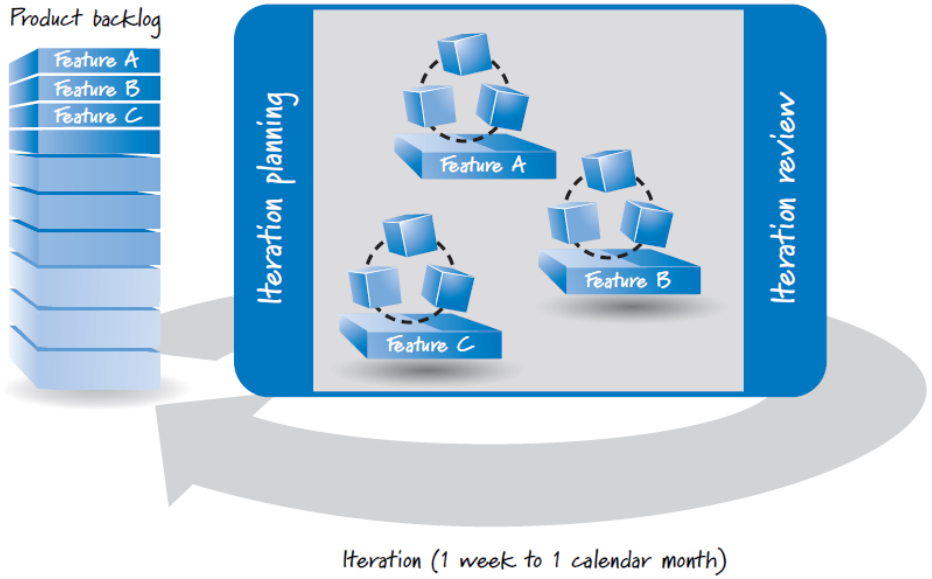
Agenda

- What is Scrum?
- Scrum Practices
  - Scrum Roles
  - Scrum Activities and Artifacts
- Scrum Values
- Scrum Principles

What is Scrum?

What is SCRUM?

- An agile method of development, in which the work itself is performed in short, timeboxed iterations, ranging from a week to a calendar month in length.
- You begin by creating a product backlog—a prioritized list of the features and other capabilities needed to develop a successful product.



SCRUM Framework

- Scrum is not a standardized process where you methodically follow a series of sequential steps that are guaranteed to produce, on time and on budget, a high-quality product that delights customers.
- The Scrum framework is based on a set of values, principles, and practices that provide the foundation to which your organization will add its unique implementation of relevant engineering practices and your specific approaches for realizing the Scrum practices.
- The result will be a version of Scrum that is uniquely yours.

SCRUM Practices

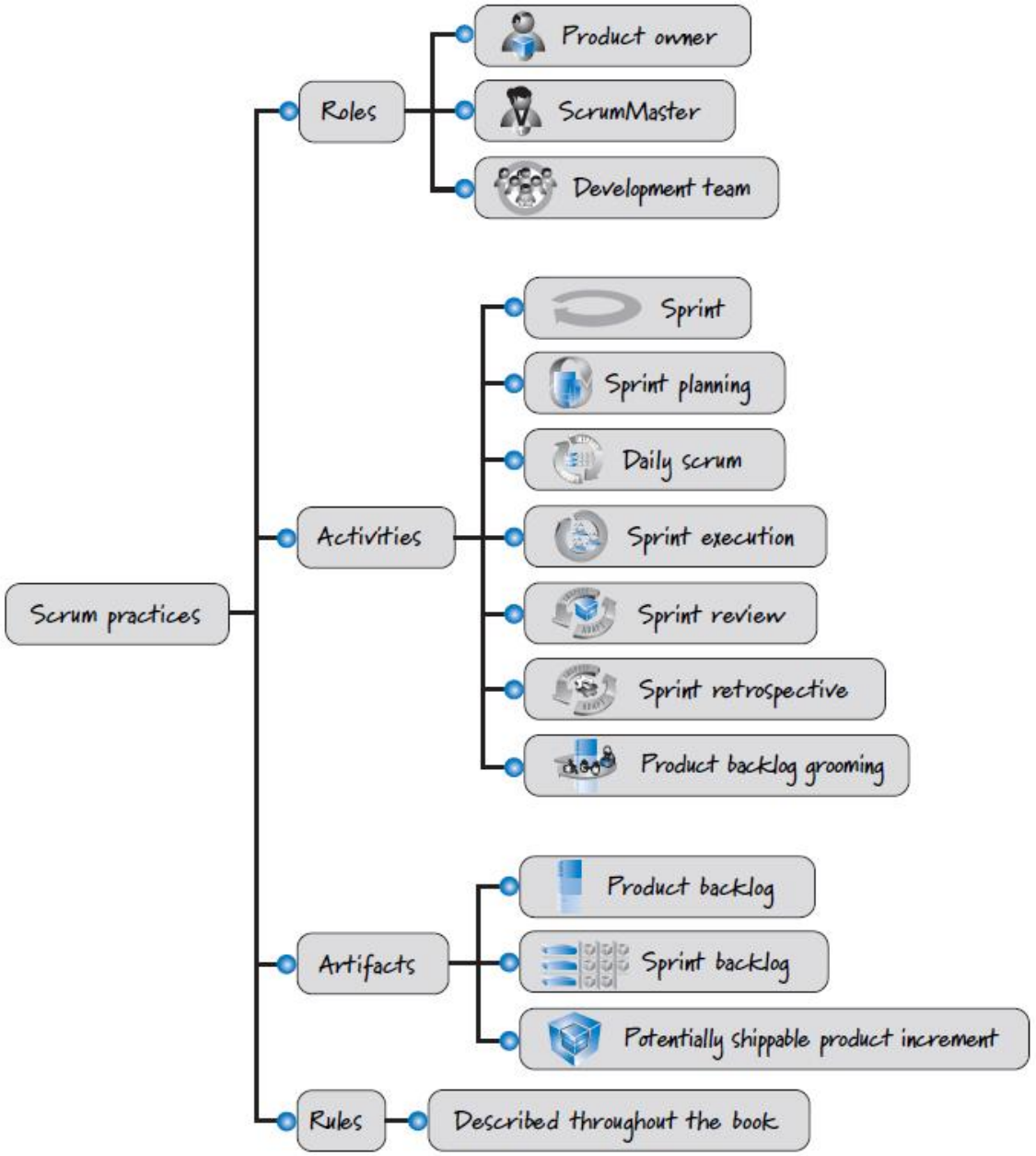
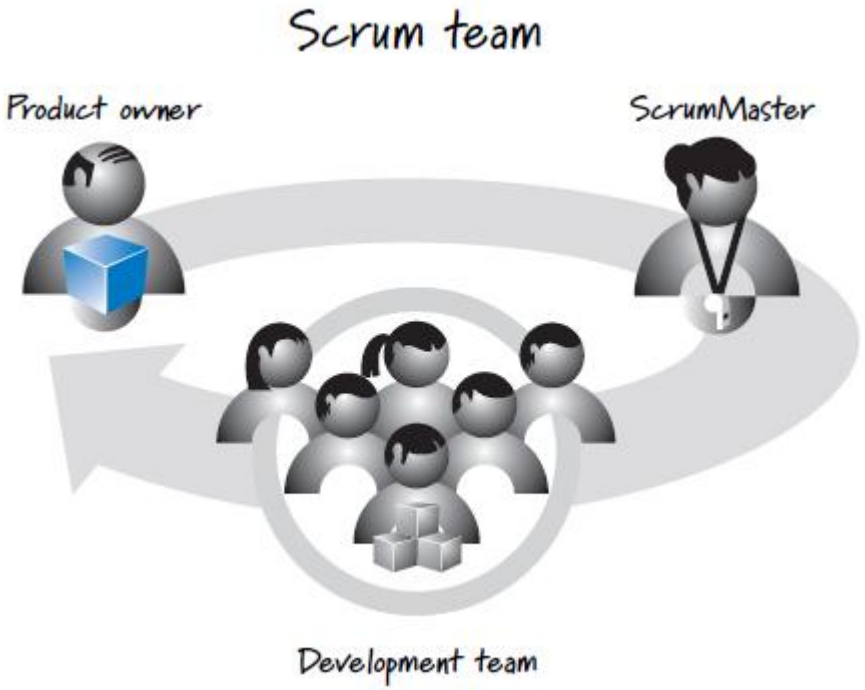


FIGURE 2.1 Scrum practices

Scrum Roles

Scrum Roles

- Scrum development efforts consist of one or more Scrum teams, each made up of three Scrum roles: product owner, ScrumMaster, and the development team



Scrum Roles: Product Owner

- The product owner is the empowered central point of product leadership.
- He is the single authority responsible for deciding which features and functionality to build and the order in which to build them.
- The product owner maintains and communicates to all other participants a clear vision of what the Scrum team is trying to achieve.
- As such, the product owner is responsible for the overall success of the solution being developed or maintained.

Scrum Roles: ScrumMaster

- The ScrumMaster helps everyone involved understand and embrace the Scrum values, principles, and practices.
- She acts as a coach, providing process leadership and helping the Scrum team and the rest of the organization develop their own high performance, organization-specific Scrum approach.
- She is also responsible for protecting the team from outside interference and takes a leadership role in removing impediments that inhibit team productivity (when the individuals themselves cannot reasonably resolve them).
- The ScrumMaster has no authority to exert control over the team, so this role is not the same as the traditional role of project manager or development manager.

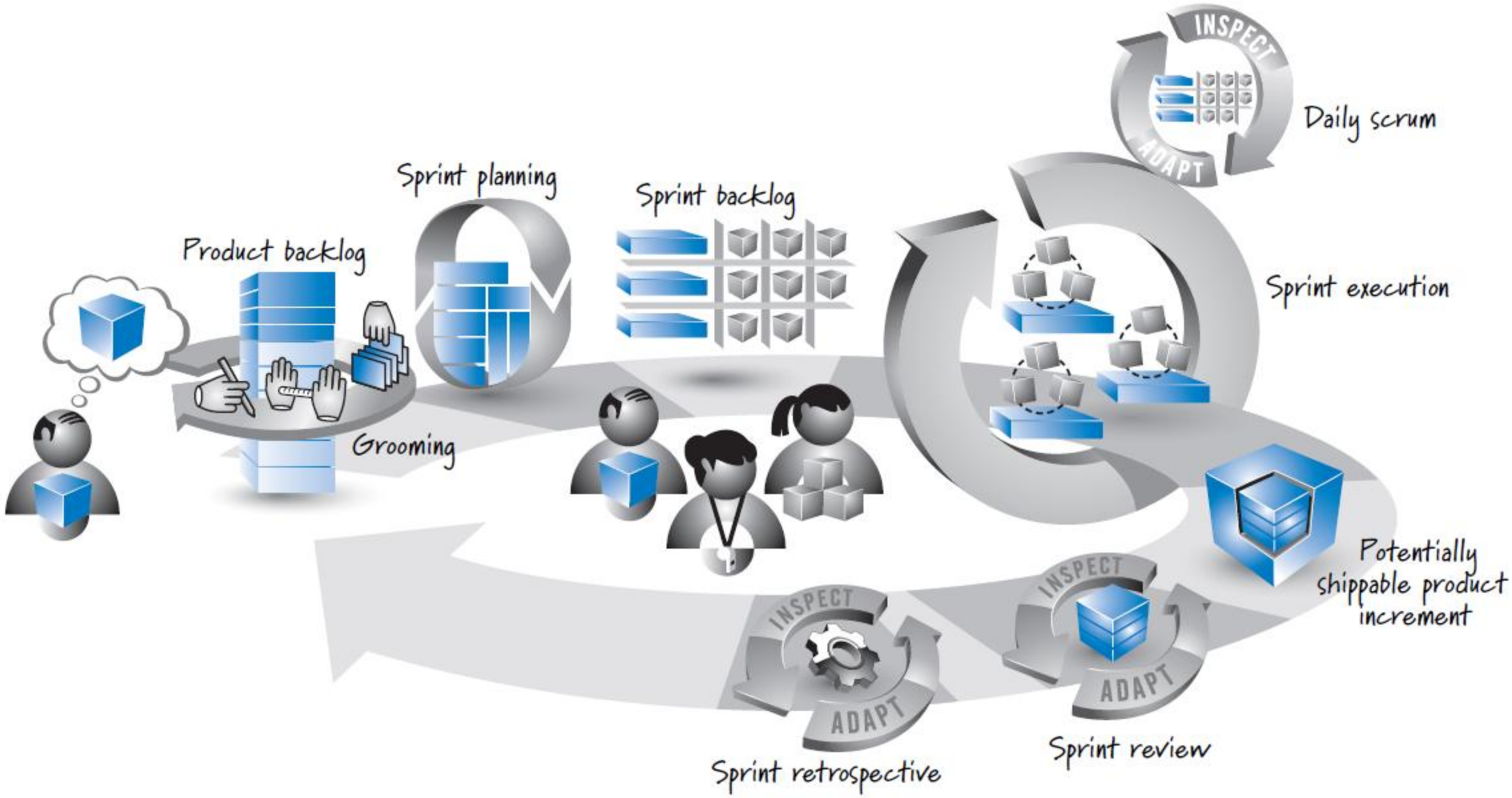
Scrum Roles: Development Team

- Traditional software development approaches discuss various job types, such as architect, programmer, tester, database administrator, UI designer, and so on.
- Scrum defines the role of a development team, which is simply a diverse, cross-functional collection of these types of people who are responsible for designing, building, and testing the desired product.
- The development team self-organizes to determine the best way to accomplish the goal set out by the product owner.

Scrum Roles: Development Team

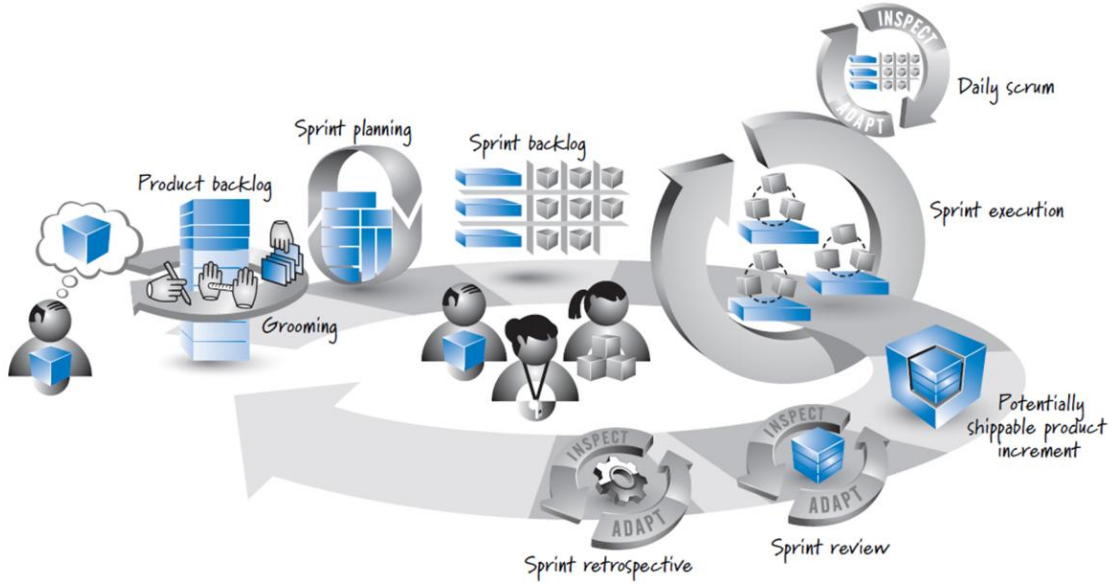
- The development team is typically five to nine people in size; its members must collectively have all of the skills needed to produce good quality, working software.
- Scrum can be used on development efforts that require much larger teams.
- However, rather than having one Scrum team with, say, 35 people, there would more likely be four or more Scrum teams, each with a development team of nine or fewer people.

Scrum Activities and Artifacts


INSPECT
ADAPT
Daily scrum
Sprint planning
Sprint backlog
Product backlog
Sprint execution
Grooming
Potentially shippable product increment
INSPECT
ADAPT
Sprint retrospective
INSPECT
ADAPT
Sprint review

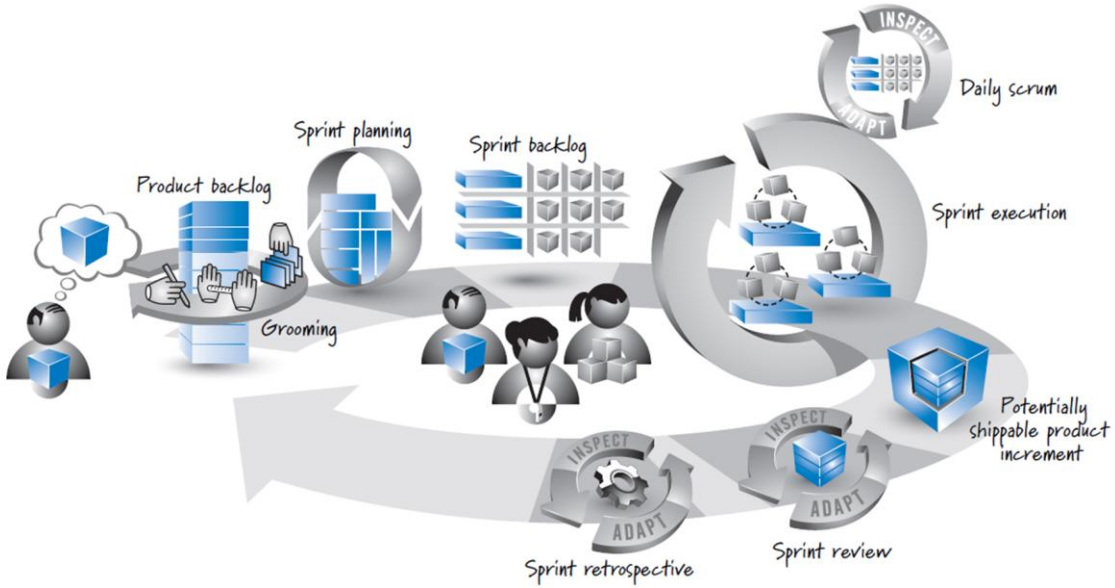
Scrum Activities and Artifacts: Product Backlog

- The product owner has a vision of what he wants to create (the big cube).
- Because the cube can be large, through an activity called grooming it is broken down into a set of features that are collected into a prioritized list called the product backlog.



Scrum Activities and Artifacts: Sprints

- A sprint starts with sprint planning, encompasses the development work during the sprint (called sprint execution), and ends with the review and retrospective.
- The sprint is represented by the large, looping arrow that dominates the centre of the figure.

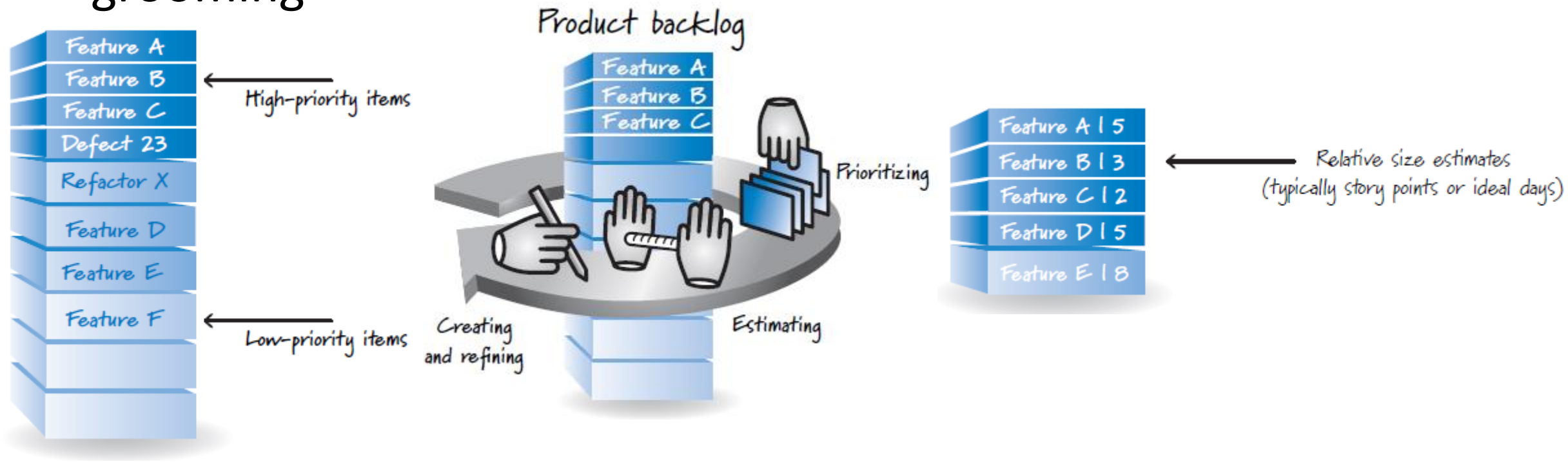


Scrum Activities and Artifacts: Sprints

- The number of items in the product backlog is likely to be more than a development team can complete in a short-duration sprint.
- For that reason, at the beginning of each sprint, the development team must determine a subset of the product backlog items it believes it can complete—an activity called sprint planning, shown just to the right of the large product backlog cube.

Scrum Activities and Artifacts: Product Backlog

- Overall, the activity of creating and refining product backlog items, estimating them, and prioritizing them is known as grooming

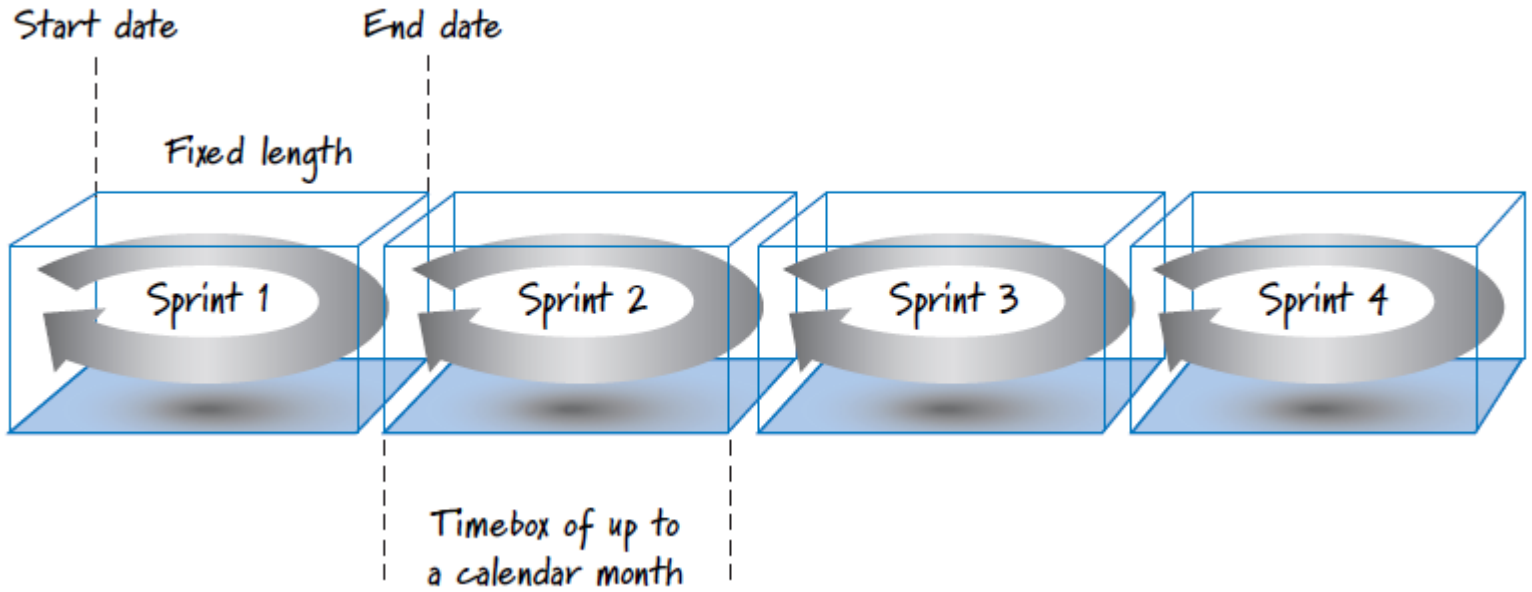


Scrum Activities and Artifacts: Product Backlog

- On new-product development the product backlog items initially are features required to meet the product owner's vision.
- For ongoing product development, the product backlog might also contain new features, changes to existing features, defects needing repair, technical improvements, and so on.

Scrum Activities and Artifacts: Sprints

- In Scrum, work is performed in iterations or cycles of up to a calendar month called sprints.



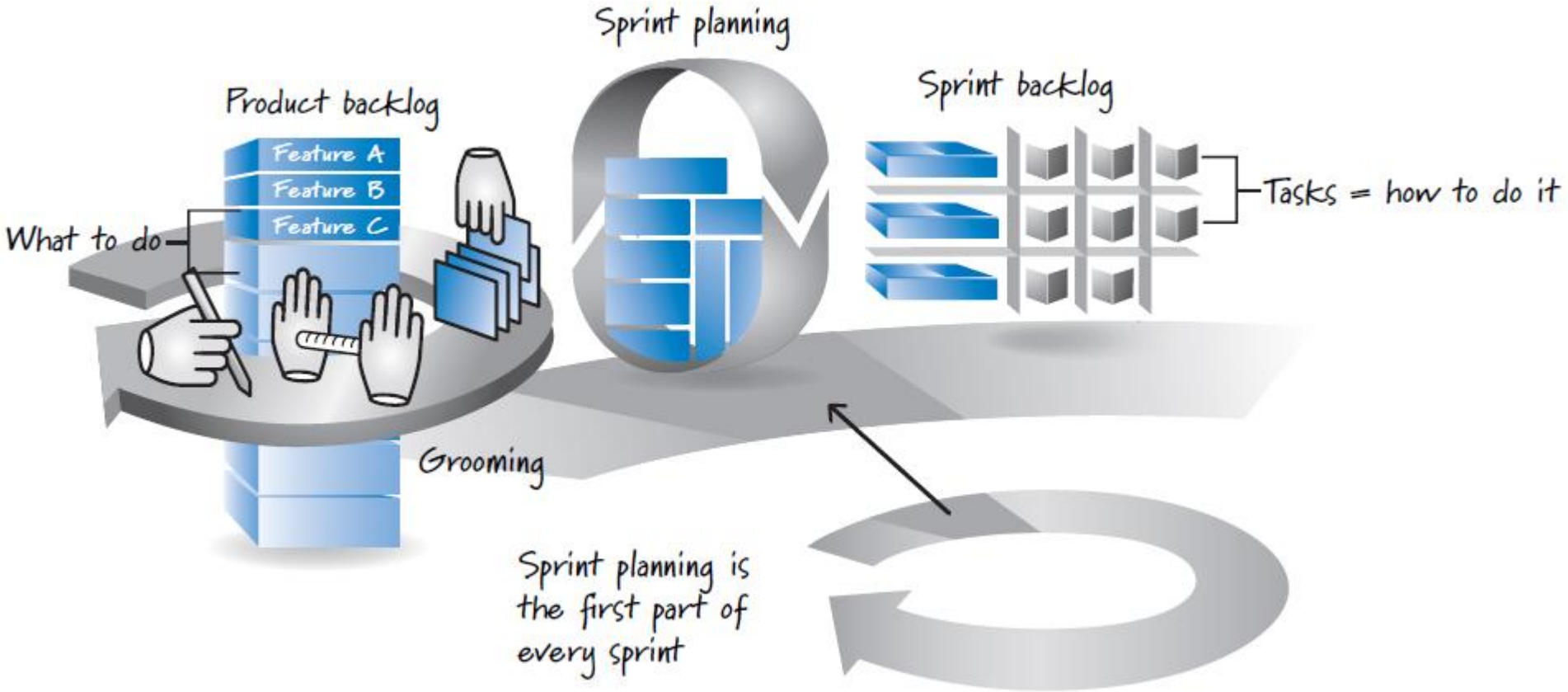
Scrum Activities and Artifacts: Sprints

- The work completed in each sprint should create something of tangible value to the customer or user.
- Sprints are timeboxed so they always have a fixed start and end date, and generally they should all be of the same duration. A new sprint immediately follows the completion of the previous sprint.

Scrum Activities and Artifacts: Sprint Planning

- A product backlog may represent many weeks or months of work, which is much more than can be completed in a single, short sprint.
- To determine the most important subset of product backlog items to build in the next sprint, the product owner, development team, and ScrumMaster perform sprint planning.

Scrum Activities and Artifacts: Sprint Planning



Scrum Activities and Artifacts: Sprint Backlog

- To acquire confidence in what it can get done, many development teams break down each targeted feature into a set of tasks. The collection of these tasks, along with their associated product backlog items, forms a second backlog called the sprint backlog.
- Any task should be scheduled to take no more than 16 hours.

Scrum Activities and Artifacts: Sprint Backlog

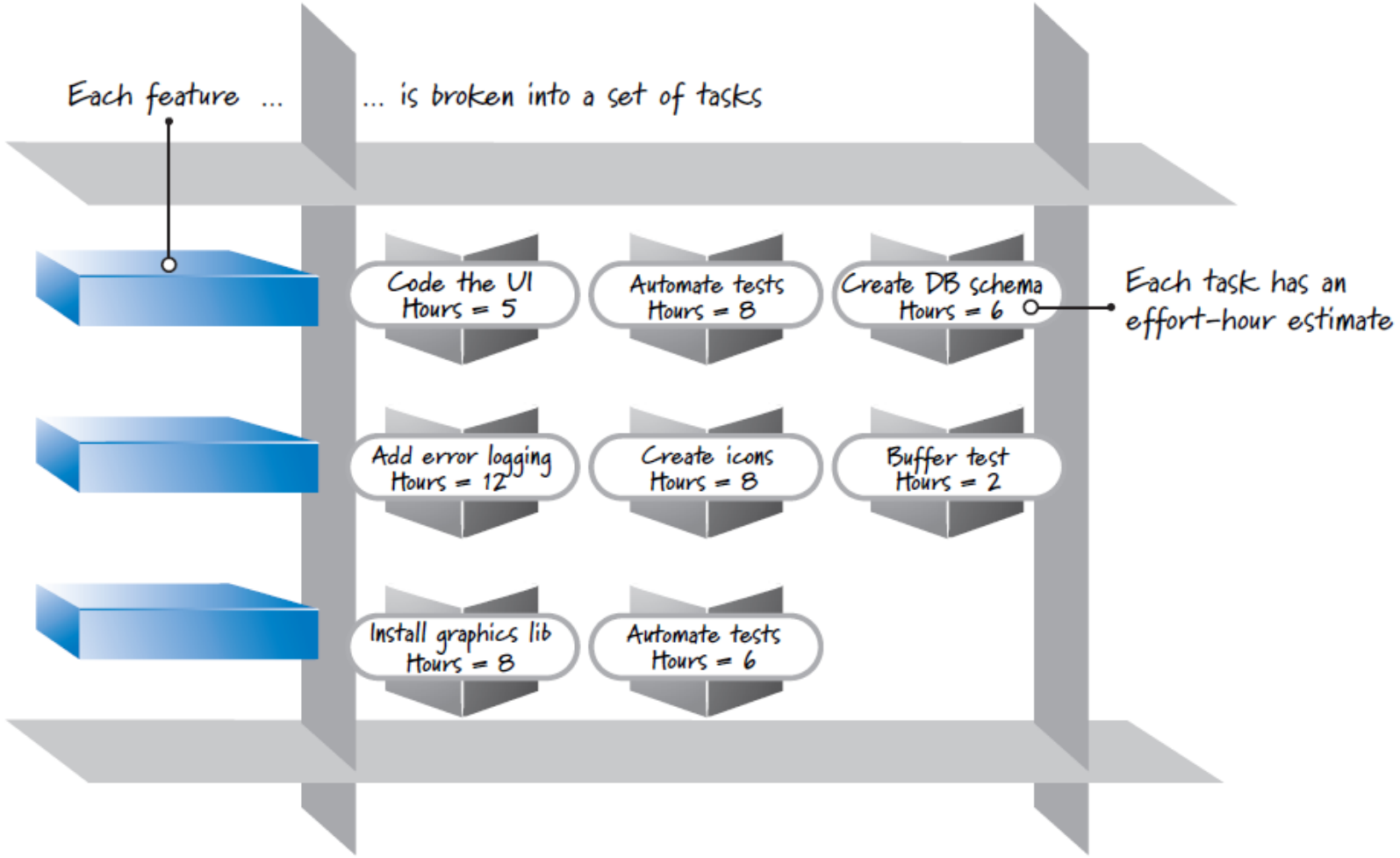


FIGURE 2.9 Sprint backlog

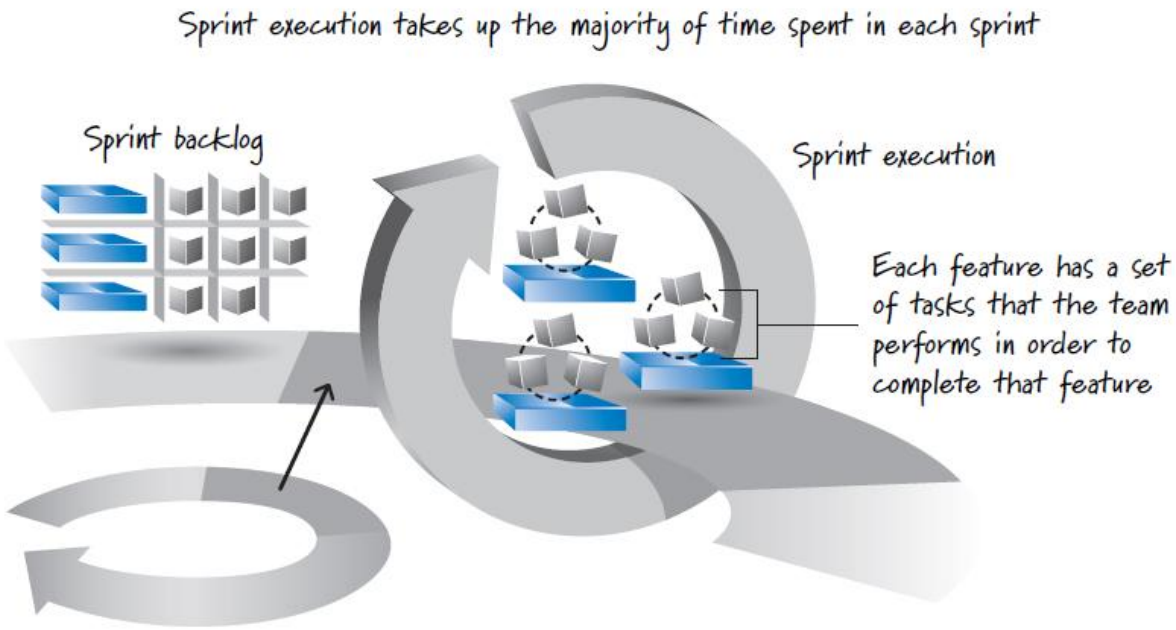
Scrum Activities and Artifacts: Sprint Backlog

				New Estimates of Effort Remaining as of Day...					
Product Backlog Item	**Sprint Task**	**Volunteer**	**Initial Estimate of Effort**	**1**	**2**	**3**	**4**	**5**	**6**
As a buyer, I want to place a book in a shopping cart	modify database		5						
	create webpage (UI)		8						
	create webpage (Javascript logic)		13						
	write automated acceptance tests		13						
	update buyer help webpage		3						
	. . .								
Improve transaction processing performance	merge DCP code and complete layer-level tests		5						
	complete machine order for pRank		8						
	change DCP and reader to use pRank http API		13						

Example of Sprint Backlog
Source: Scrum Training Institute Press

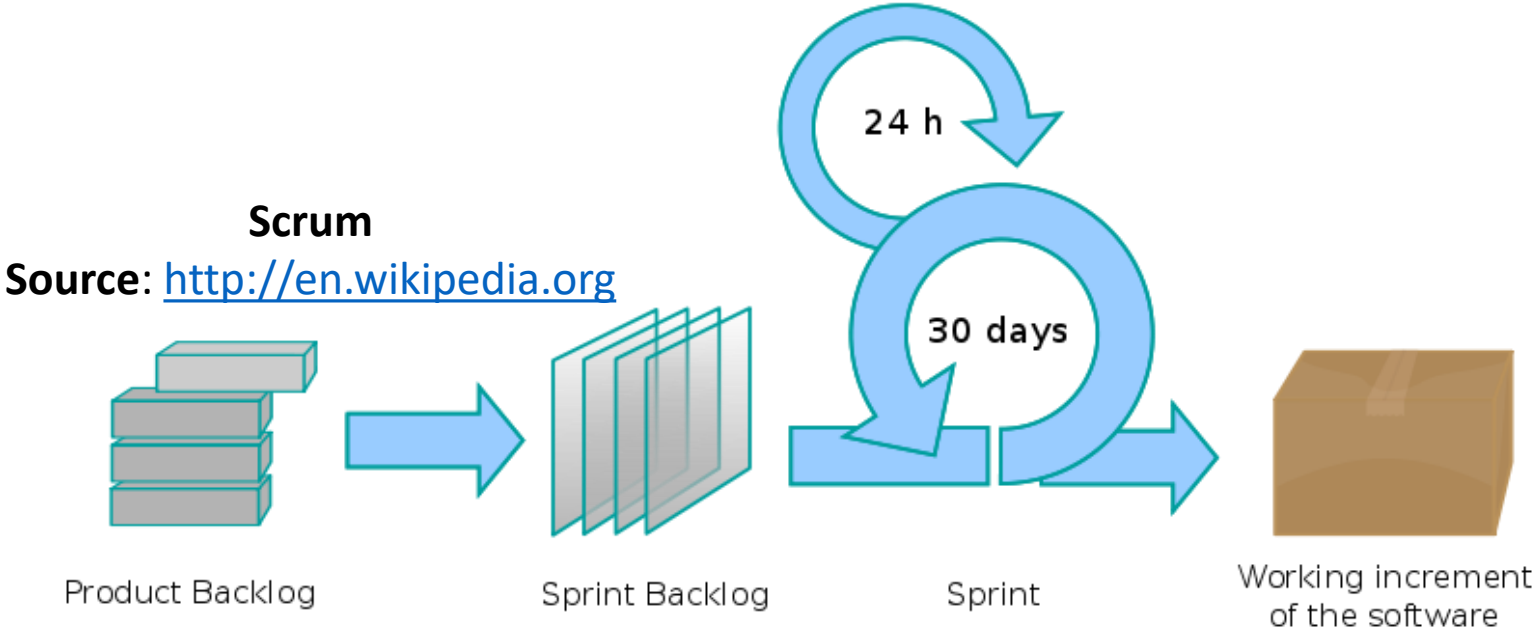
Scrum Activities and Artifacts: Sprint Execution

- Once the Scrum team finishes sprint planning and agrees on the content of the next sprint, the development team, guided by the ScrumMaster's coaching, performs all of the task-level work necessary to get the features done.



Scrum Activities and Artifacts: Sprint Execution

- **El sprint burn down chart** is a graphic that shows the amount of pending requirements in the product backlog at the beginning of each sprint.



Scrum

Source: http://en.wikipedia.org

Scrum Activities and Artifacts: Sprint Execution

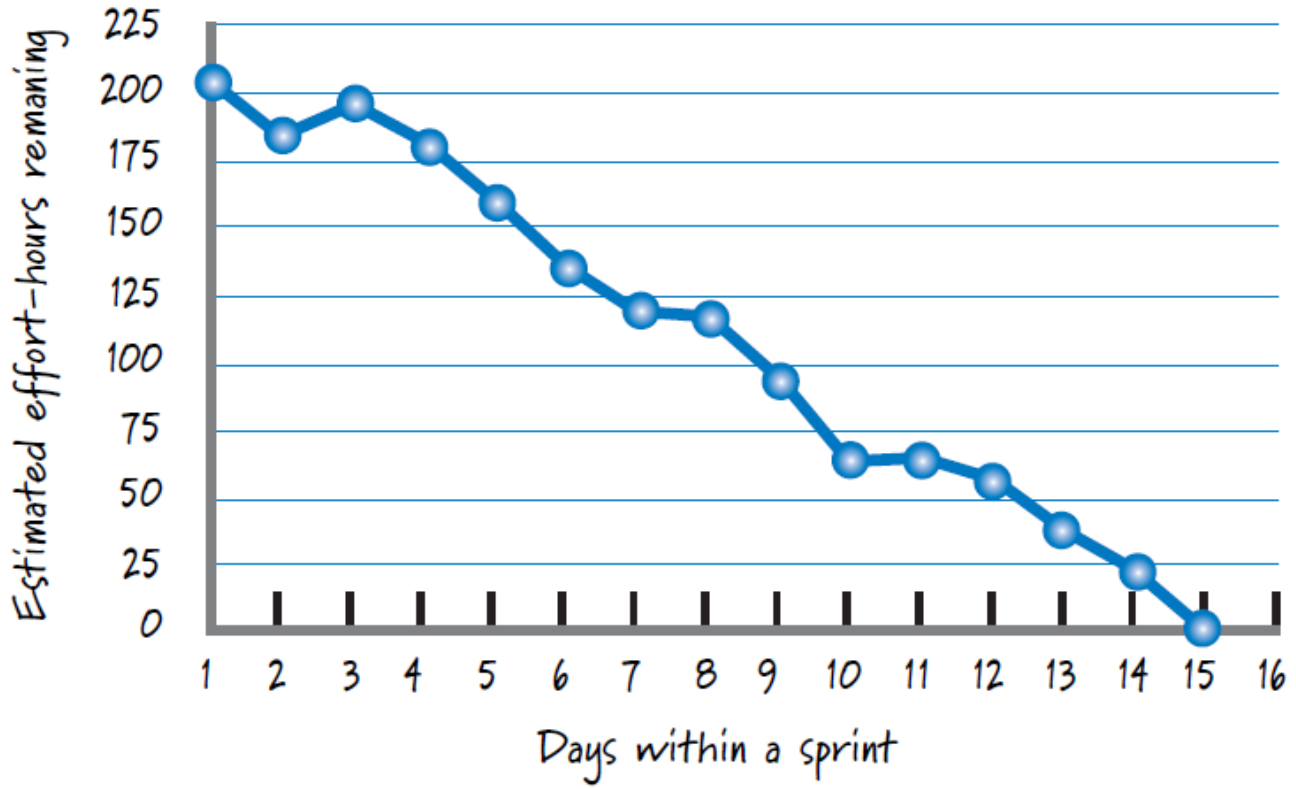
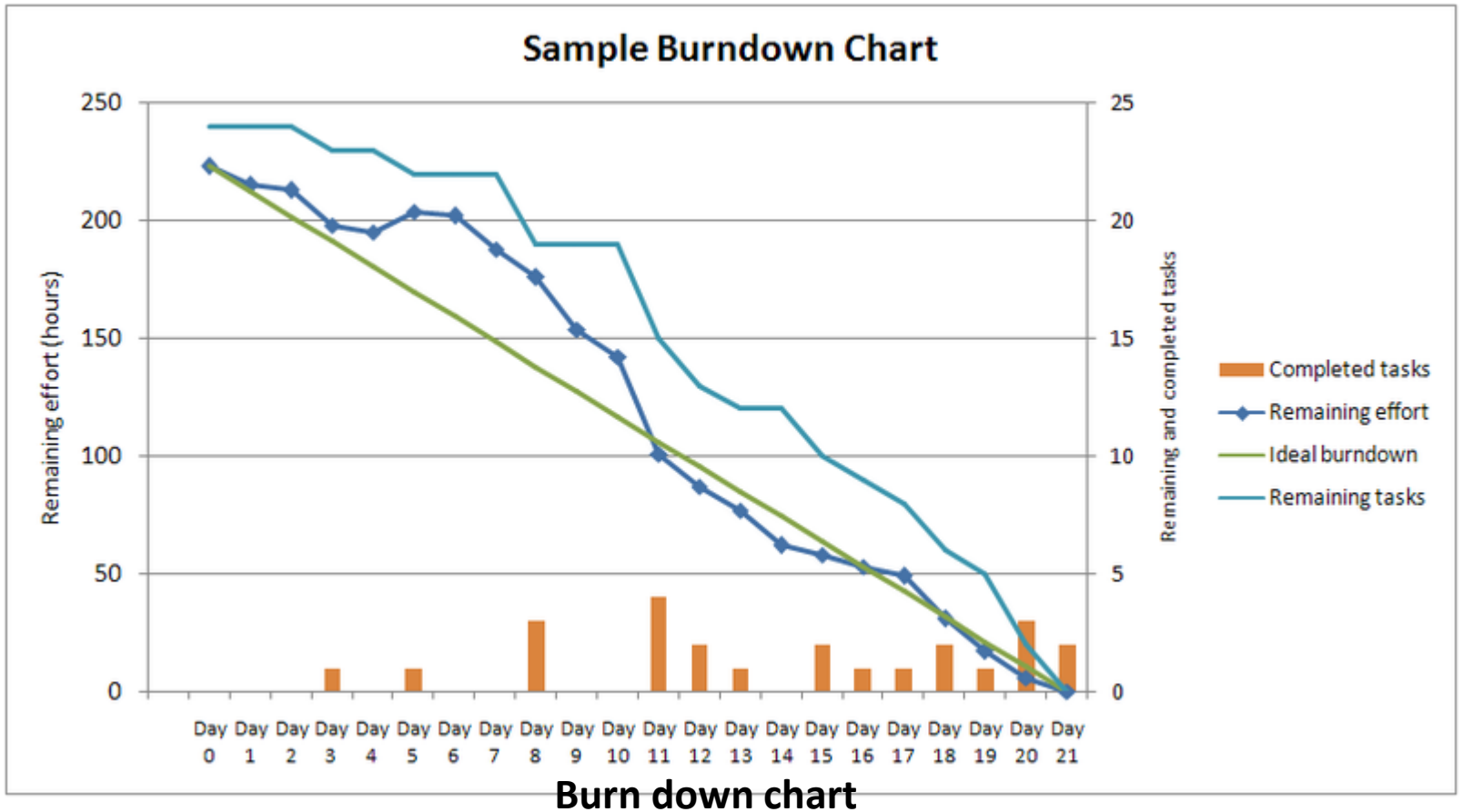


FIGURE 20.7 Sprint burndown chart

Sprint burndown chart

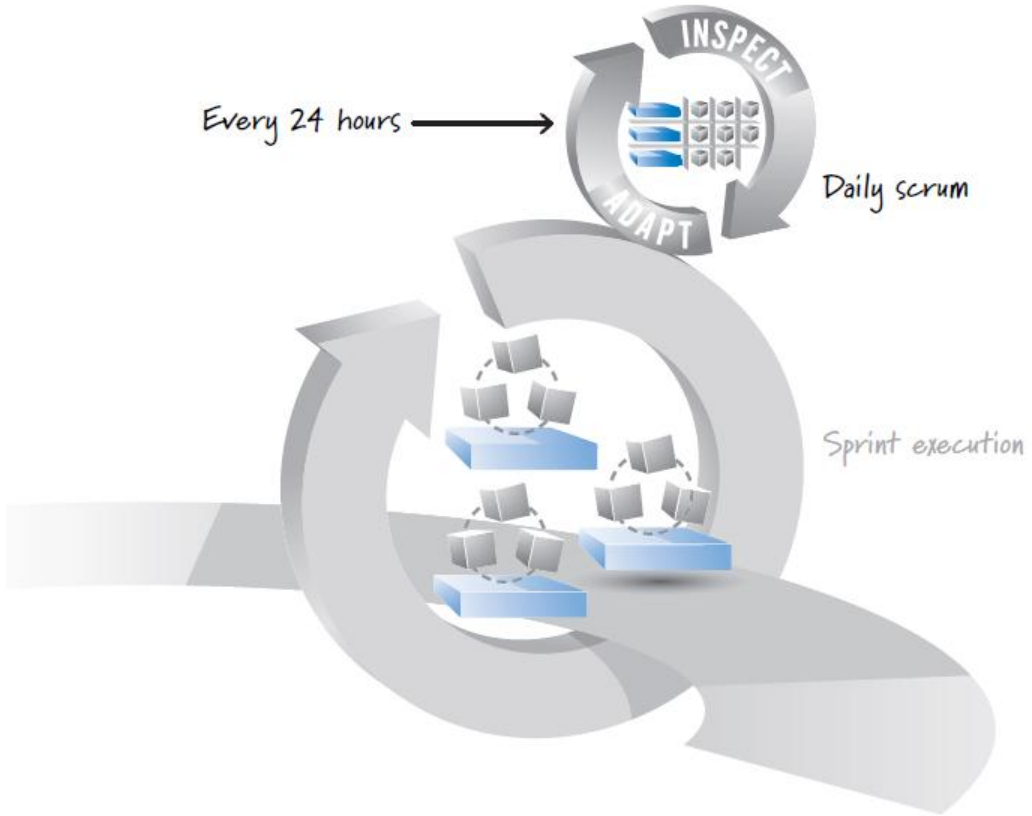


Burn down chart

Source: http://en.wikipedia.org

Scrum Activities and Artifacts: Daily Scrum

- Each day of the sprint, ideally at the same time, the development team members hold a timeboxed (15 minutes or less) daily scrum.

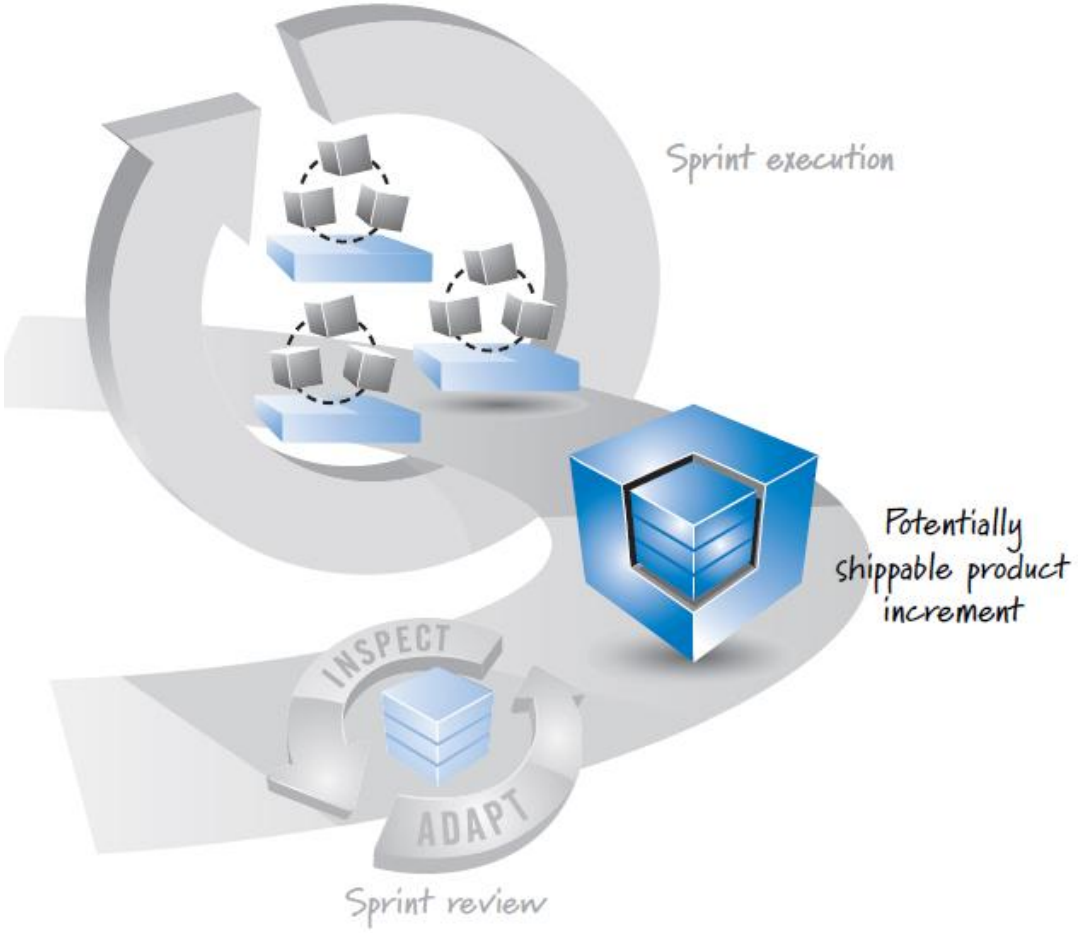


Scrum Activities and Artifacts: Daily Scrum

- Each team member taking turns answering three questions for the benefit of the other team members:
  - What did I accomplish since the last daily scrum?
  - What do I plan to work on by the next daily scrum?
  - What are the obstacles or impediments that are preventing me from making progress?
- By answering these questions, everyone understands the big picture of what is occurring, how they are progressing toward the sprint goal, any modifications they want to make to their plans for the upcoming day's work, and what issues need to be addressed.

Scrum Activities and Artifacts: Done

- We refer to the sprint results as a potentially shippable product increment.



Scrum Activities and Artifacts: Sprint Review

- At the end of the sprint there are two additional inspect-and-adapt activities. One is called the sprint review.
- The goal of this activity is to inspect and adapt the product that is being built.
- Critical to this activity is the conversation that takes place among its participants, which include the Scrum team, stakeholders, sponsors, customers, and interested members of other teams.

Scrum Activities and Artifacts: Sprint Review

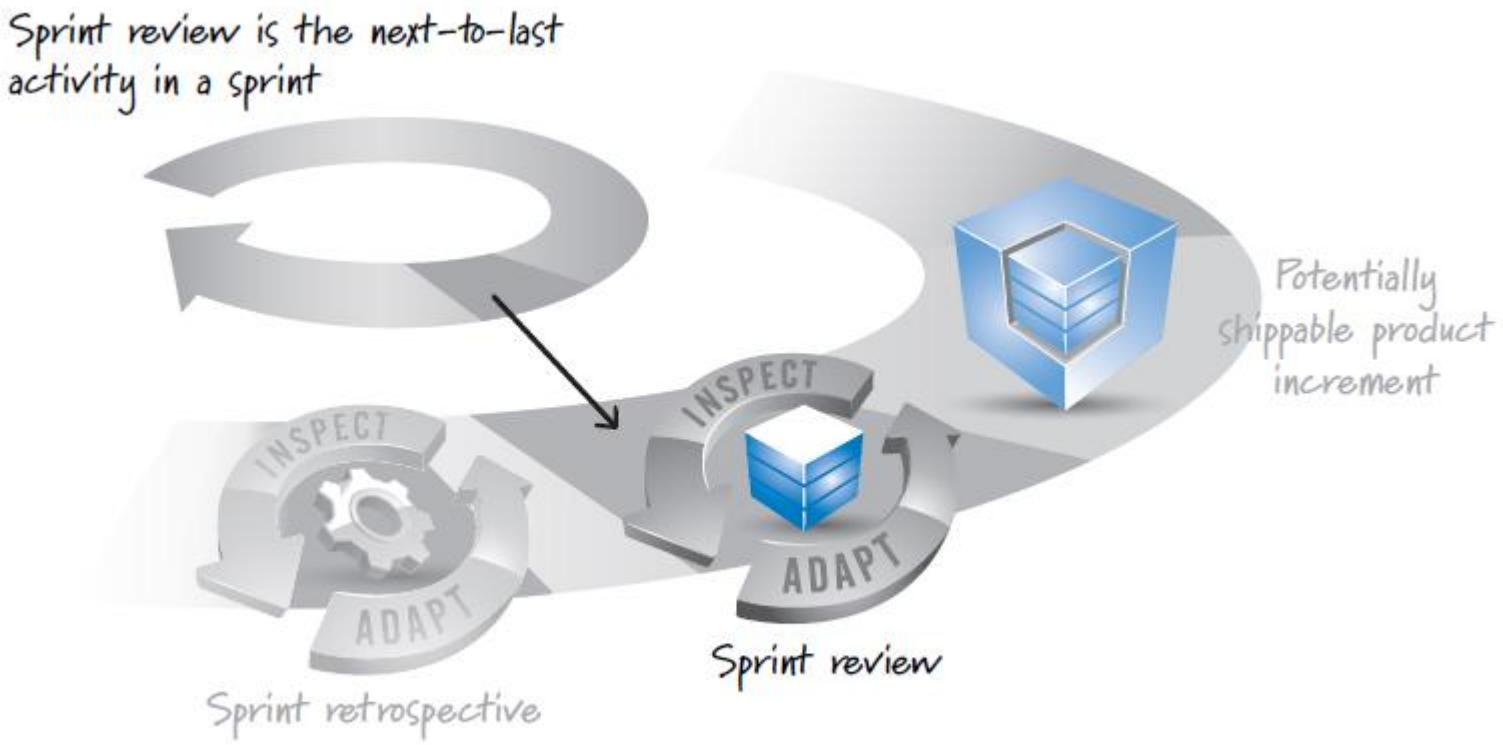
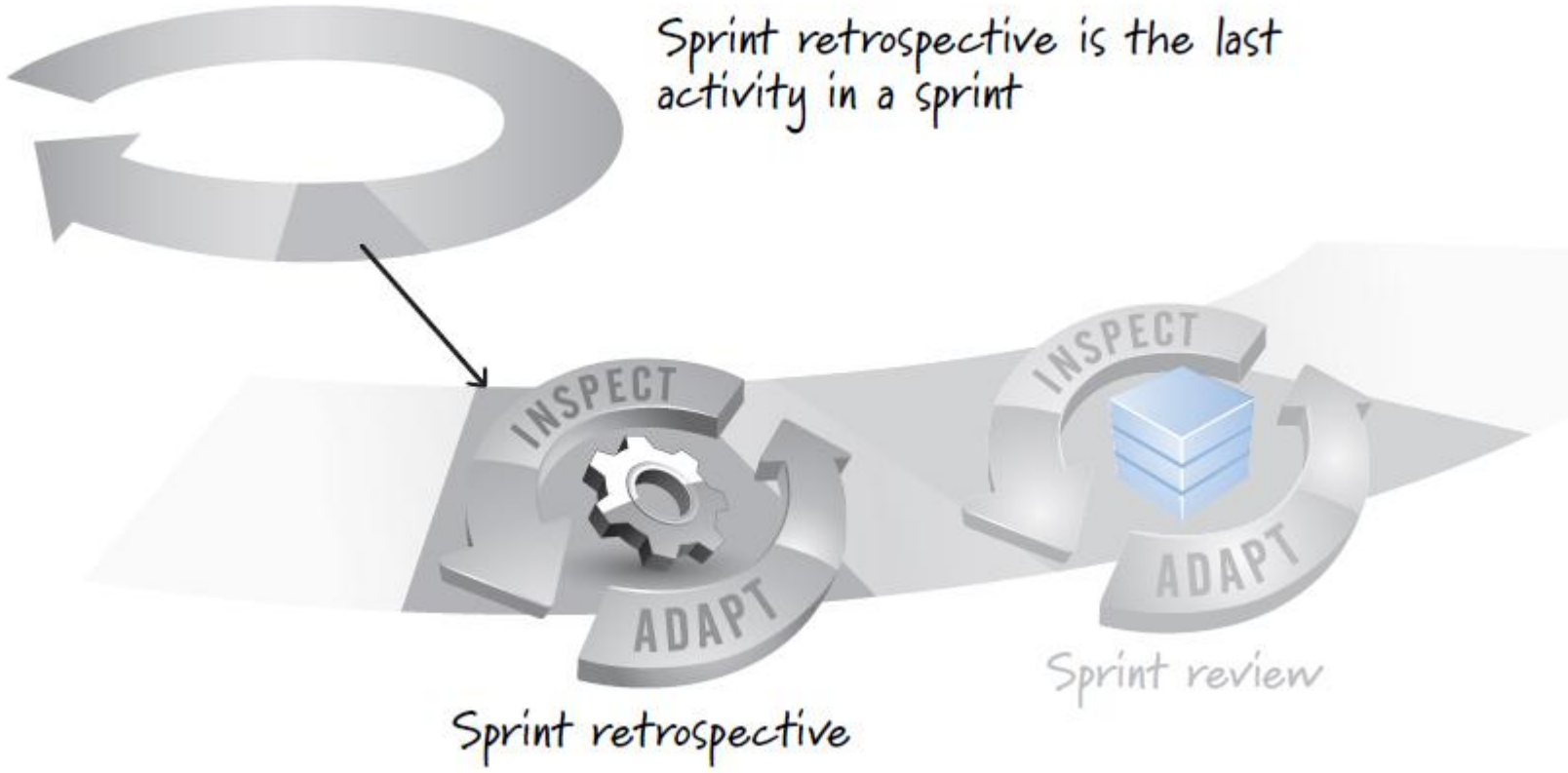


FIGURE 2.13 Sprint review

Scrum Activities and Artifacts: Sprint Retrospective

- The second inspect-and-adapt activity at the end of the sprint is the sprint retrospective.
- This activity frequently occurs after the sprint review and before the next sprint planning.
- Whereas the sprint review is a time to inspect and adapt the product, the sprint retrospective is an opportunity to inspect and adapt the process.
- During the sprint retrospective the development team, ScrumMaster, and product owner come together to discuss what is and is not working with Scrum and associated technical practices.

Scrum Activities and Artifacts: Sprint Retrospective

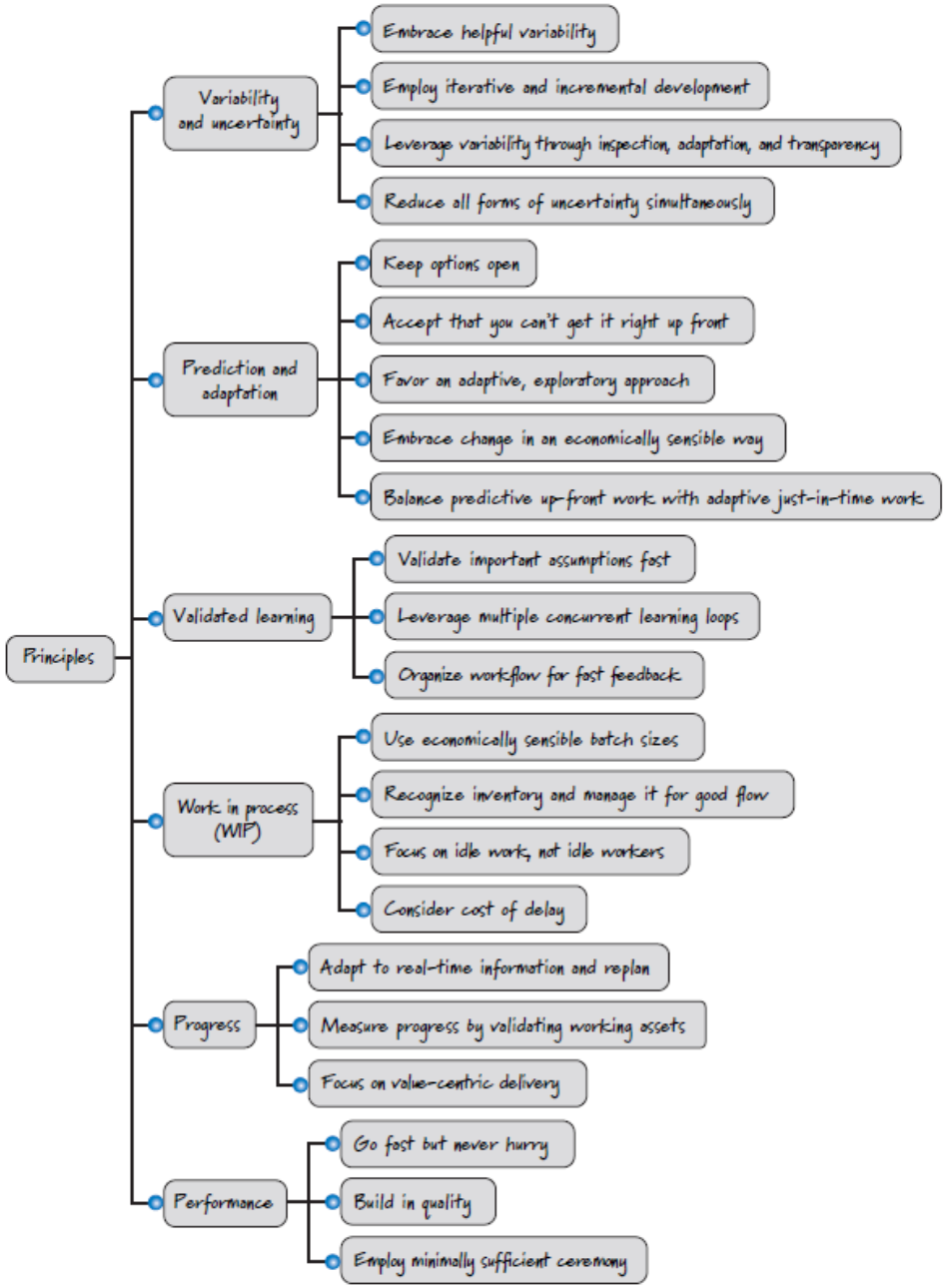


Scrum Values

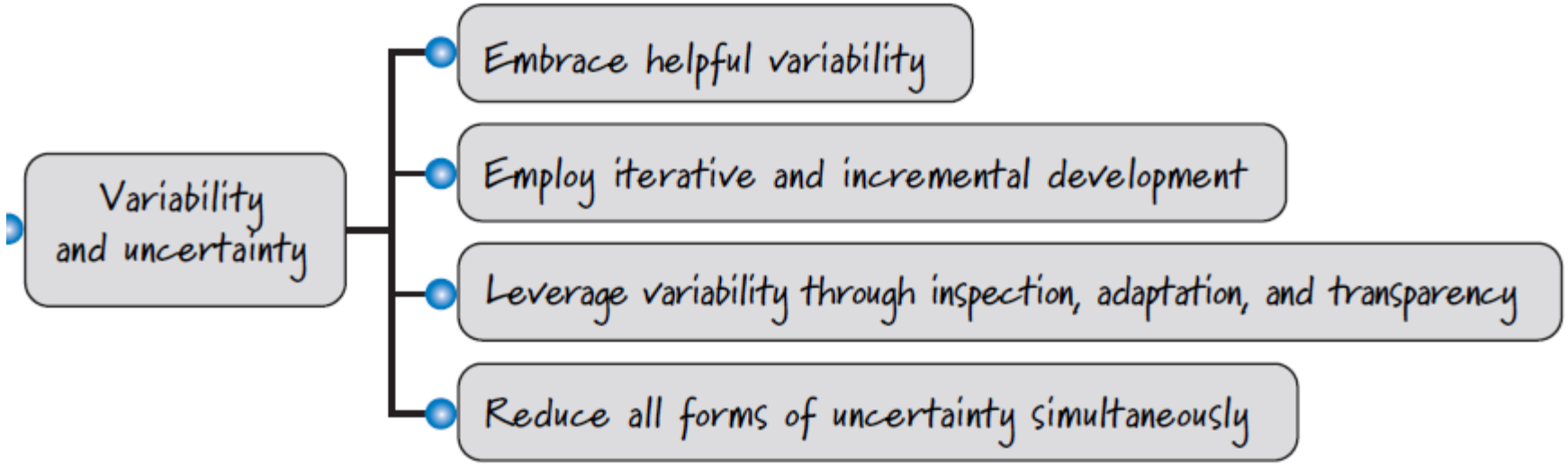
SCRUM Values

- Honesty
- Openness
- Courage
- Respect
- Focus
- Trust
- Empowerment
- Collaboration

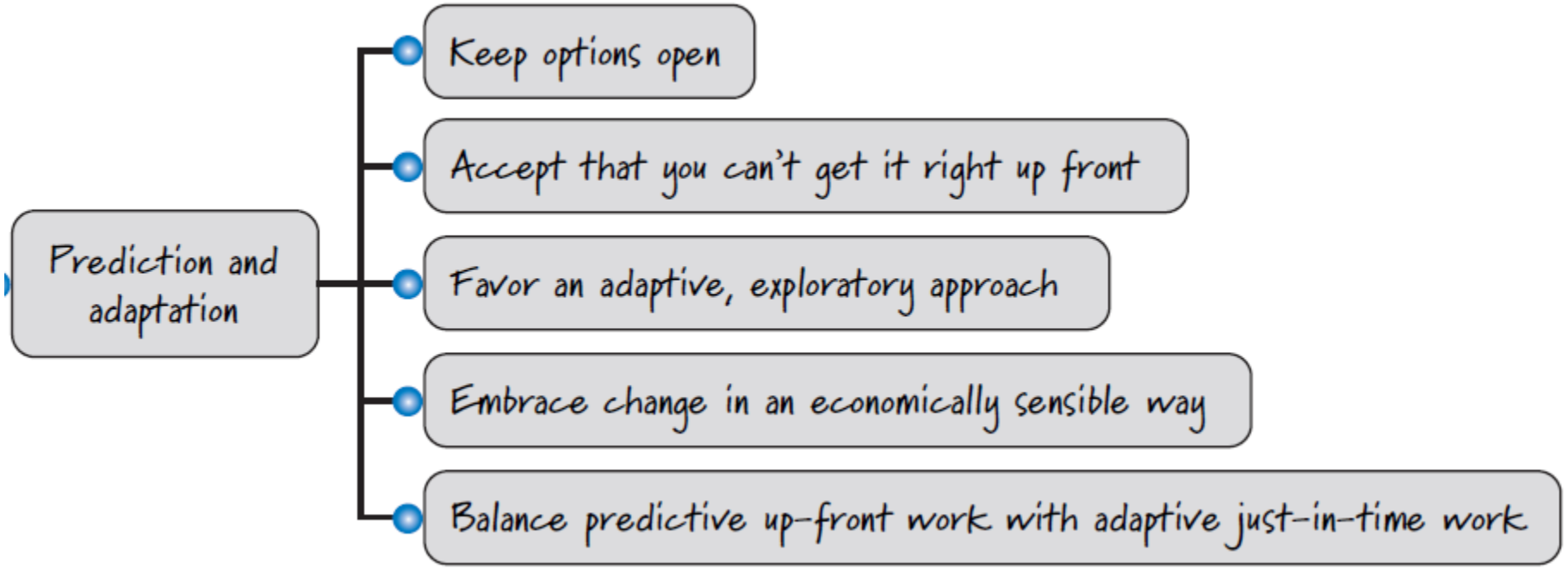
Scrum Principles


Principles
Variability and uncertainty
Embrace helpful variability
Employ iterative and incremental development
Leverage variability through inspection, adaptation, and transparency
Reduce all forms of uncertainty simultaneously
Prediction and adaptation
Keep options open
Accept that you can't get it right up front
Favor an adaptive, exploratory approach
Embrace change in an economically sensible way
Balance predictive up-front work with adaptive just-in-time work
Validated learning
Validate important assumptions fast
Leverage multiple concurrent learning loops
Organize workflow for fast feedback
Work in process (WIP)
Use economically sensible batch sizes
Recognize inventory and manage it for good flow
Focus on idle work, not idle workers
Consider cost of delay
Progress
Adapt to real-time information and replan
Measure progress by validating working assets
Focus on value-centric delivery
Performance
Go fast but never hurry
Build in quality
Employ minimally sufficient ceremony

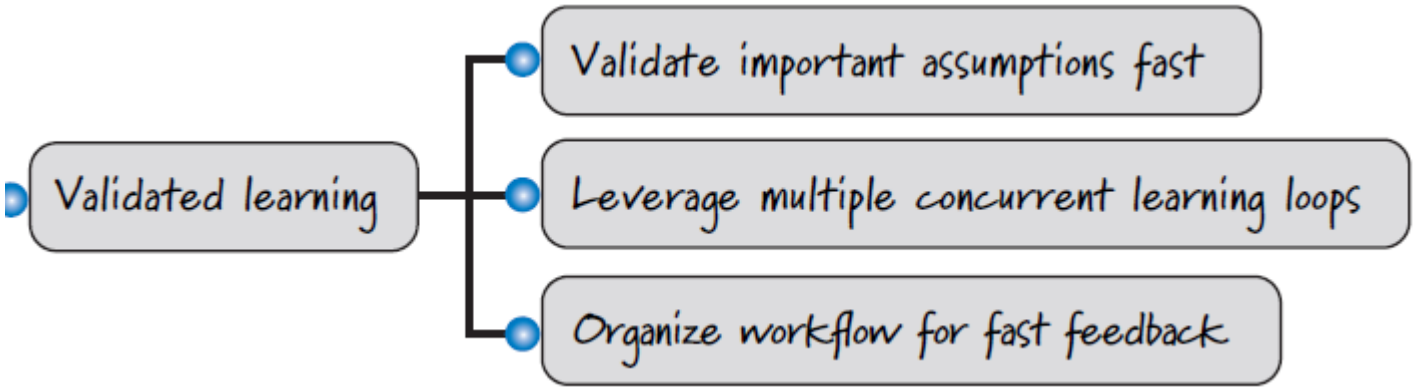
Scrum Principles



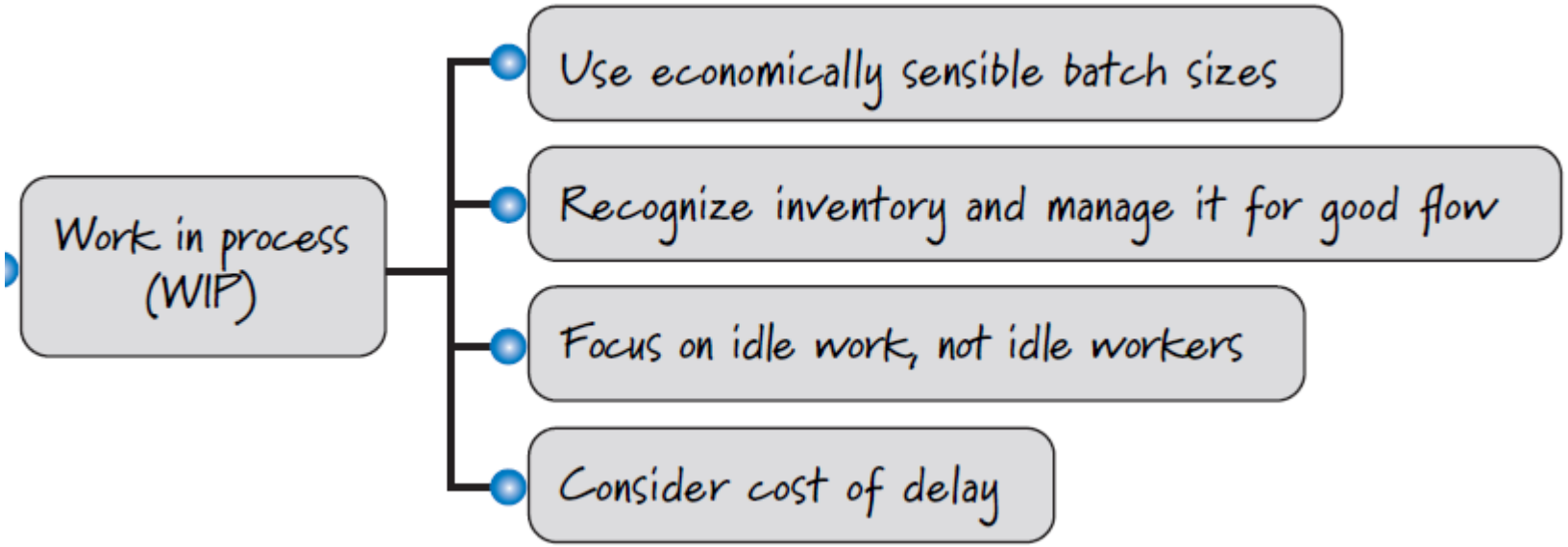
Scrum Principles



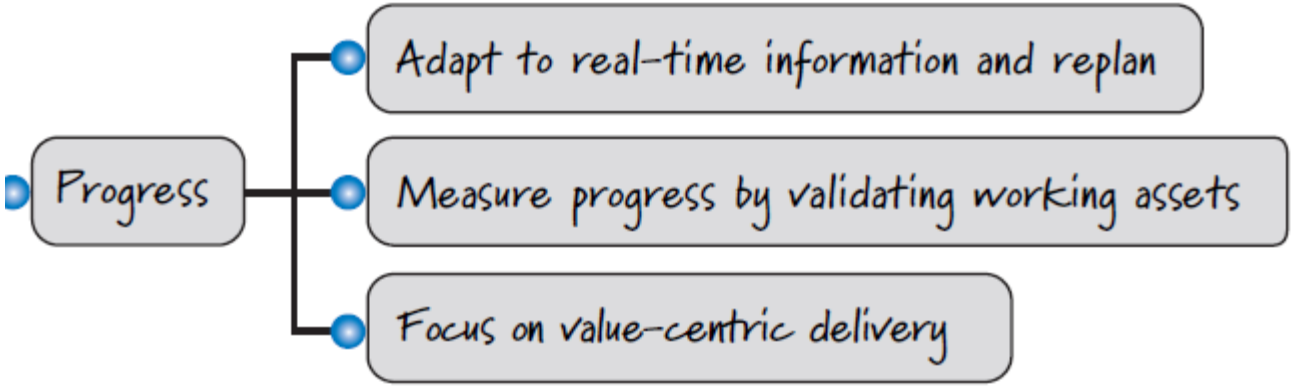
Scrum Principles



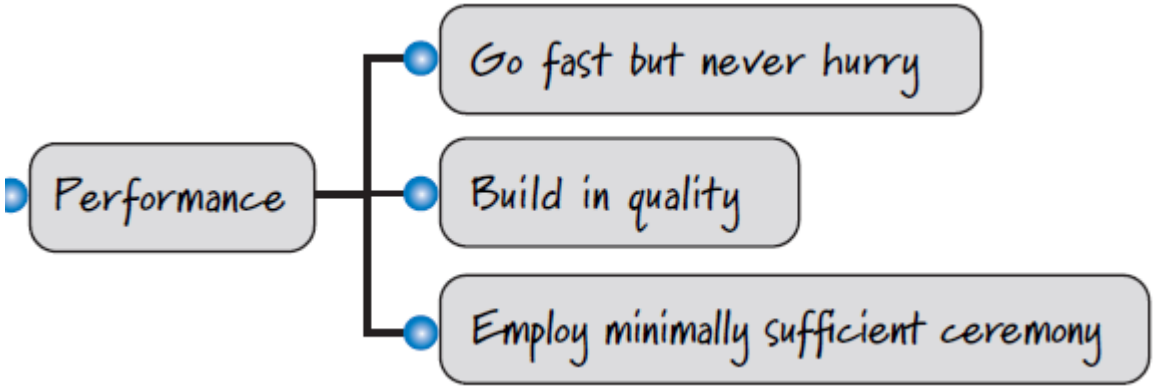
Scrum Principles



Scrum Principles



Scrum Principles



Benefits and Limitations of Scrum

Benefits of adopting Scrum

- The delivery of a potentially shippable (functional) product at the end of each sprint.
- The product is divided in fully understandable and manageable subsets.
- Unstable requirements are promptly identified.
- All the team knows the status of the project.
- A relation of trust is built between customer and developers.

Limitations of adopting Scrum

- More challenging to apply in big projects.
- Requires a team with transversal skills.
- Assumes a very active customer who is continuously involved in the development.
- Some documentation required by diverse international standards is not generated.

Next On

Take Away Points

- What is Scrum?
- Which are its activities?
- Which are its values?
- Can you describe each of the Scrum artifacts?
- Which are the Scrum roles?
- Which are the Scrum principles?
- What are the benefits and limitations of Scrum?

Further Reading

- Kenneth Rubin, “Essential Scrum: A Practical Guide to the Most Popular Agile Process”
- International Scrum Institute, “The Scrum Framework”
- Mike Cohn, “User Stories Applied for Agile Software Development”

Next Lecture

- Requirements Engineering